



FUTURE
INTERNS

Phishing Detection & Awareness Report

Cyber Security Internship – Task 2 (2026)

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Executive Summary

This report presents the findings of a phishing email analysis conducted as part of a cybersecurity internship project. Three email samples were examined using email header analysis tools to identify spoofing attempts and authentication failures. Two emails were classified as phishing due to SPF, DKIM, and DMARC failures along with domain impersonation techniques. One email was verified as legitimate based on proper authentication mechanisms. The findings emphasize the importance of email security controls and user awareness in preventing phishing attacks within organizations.

Objectives:

- Analyze phishing email samples.
- Identify common phishing indicators.
- Classify emails based on risk level.
- Explain phishing techniques in simple language.
- Provide prevention and awareness guidelines.

Tools Used:

- Google Admin Toolbox – Message Header Analyzer
- Browser Inspection Tools
- Microsoft Word for documentation
- GitHub for repository submission

Methodology:

1. Collected phishing and legitimate email samples.
2. Extracted full email headers.
3. Analyzed SPF, DKIM, and DMARC authentication results.
4. Verified sender domain legitimacy.
5. Inspected mail routing path and sending IP.
6. Identified phishing indicators.
7. Classified emails based on risk severity.

Email Analysis:

Email 1 – Kraken Security (Phishing)

Subject: Urgent Account Verification Required

Authentication Results: SPF – Fail, DKIM – None, DMARC – Fail

Findings: The email failed all authentication checks and originated from an unknown IP address. The message used urgency-based language to pressure the recipient into immediate action.

Classification: PHISHING – High Risk

Email 2 – HDFC Support (Phishing)

Subject: Important: Action Required for Your Bank Account

Authentication Results: SPF – Fail, DKIM – None, DMARC – Fail

Findings: The sender domain (hdfc-supports.com) impersonates a legitimate banking institution. Authentication failures indicate spoofing. The email uses fear tactics to manipulate the user.

Classification: PHISHING – High Risk

Email 3 – Amazon Order Confirmation (Legitimate)

Subject: Your Order Confirmation

Authentication Results: SPF – Pass, DKIM – Pass, DMARC – Pass

Findings: All authentication mechanisms passed successfully. The email originated from a legitimate Amazon mail server.

Classification: SAFE – Low Risk

Risk Classification Summary

Email	SPF	DKIM	DMARC	Risk Level	Classification
Kraken	Fail	None	Fail	High	Phishing
HDFC	Fail	None	Fail	High	Phishing
Amazon	Pass	Pass	Pass	Low	Safe

Common Phishing Techniques Observed

- Domain Spoofing and Typosquatting
- Urgency and Fear-Based Messaging
- Authentication Failures (SPF, DKIM, DMARC)
- Impersonation of Trusted Brands

Prevention & Awareness Guidelines

Technical Controls

- Enable SPF, DKIM, and DMARC policies.
- Use email filtering and anti-spam gateways.
- Implement Multi-Factor Authentication (MFA).

Employee Awareness

- Verify sender domain carefully.
- Avoid clicking suspicious links.
- Report suspicious emails to IT/Security team.

Organizational Practices

- Conduct regular phishing awareness training.
- Establish clear incident response procedures.
- Monitor domain spoofing attempts.

Conclusion

The analysis demonstrates how phishing emails exploit authentication weaknesses and social engineering techniques to deceive users. By implementing strong email security controls and conducting user awareness training, organizations can significantly reduce phishing risks.